CENG 101 – Algorithms and Programming I

ASSIGNMENT

In this assignment, students are expected to write a C program that performs several checks on a user-provided Turkish ID Number (T.C. Kimlik Numarası), then conducts a special ID number validation, and finally executes either a vowel-counting or a season-determination operation based on specific conditions.

The program should consist of the following functions, each carrying out the tasks described below:

`main`

The program must request the user to enter an ID number. If the user enters an invalid ID number, the program should repeatedly prompt them until a valid one is provided. For an ID number to be considered valid, all of the following four conditions must be satisfied:

- the ID number must contain exactly 11 digits.
- the first digit must not be 0.
- the last digit must be an even number.
- the ones digit of the sum of the first 10 digits must be equal to the last digit of the ID number.

If any of these conditions are not met, the ID number is considered invalid, and the user should be informed and prompted again. The user must also be allowed to exit the program at any time instead of entering another ID number. When the user prefers exiting, the program must terminate.

Once a valid number is provided, the main function must call the `specialityChecker` function. During this call, the `main` function should pass the first, middle (sixth), and last digits of the ID number to `specialityChecker` function as parameters.

`specialityChecker`

This function checks whether the user-entered valid ID number is a special ID number. The condition for being considered special is:

- The ones digit of the product (*sum of the first and last digits*) × (*positive difference of the first and last digits*) must match the middle (sixth) digit of the ID number.

After performing this check, the `specialityChecker` function must return the result to the `main` function. Based on this result, the `main` function should display an appropriate message to the user in order to inform the user about whether the ID is special or not. If the ID is special, the `main` function should call the `seasonFinder` function; otherwise, it should call the `vowelCounter` function.

`seasonFinder`

This function should ask the user to enter their birth month as a number between 1 and 12. The user is allowed up to 3 attempts. After each invalid entry, the program must inform the user of the remaining attempts. If the user makes three invalid attempts, the program should display an informative message and terminate.

If a valid month is entered, the function must determine the corresponding season according to the following list:

- December, January, February → Winter
- March, April, May → Spring
- June, July, August → Summer
- September, October, November → Autumn

After displaying the season result to the user, the program should terminate.

`vowelCounter`

This function must ask the user to enter their full name. It should then count the number of vowels in the name (*case-insensitive*), considering the English vowel set: a, e, i, o, u. The result should be printed on the screen. After completing the operation, the program should terminate.

Technical Requirements

- The project must be completed using only the concepts taught in Lec01–Lec09. No additional features beyond these should be used. For example, data structures such as arrays, string variables, and string manipulation functions must not be used.
- The statements `switch`, `if-else`, `while` (`do…while`), and `for` must each be used at least once in a meaningful way.
- Students may define additional functions if they wish; however, the four functions listed above are mandatory and must operate exactly as described.
- You may use functions from the Math library whenever you consider them necessary.
- Functions and variables name should be meaningful.
- The program should behave as closely as possible to the sample screenshots provided. These screenshots may be used for guidance and for testing your program.
- The code should be briefly explained using comment lines in English, where appropriate.
- The code must be compliable and executable. The program that is not executable is going to be graded 0.

Submission Requirements:

- The project must be completed in groups of 2 or 3 students.
  - A list of groups is attached to this file. No changes will be made to the groups under any circumstances, regardless of the reason.
- The project should be submitted as a single .c file, and only one member of the group should submit it via email before the deadline.
- **File Naming Format:** `ceng101_group_groupNo_hw.c`
  - *example:* `ceng101_group_92_hw.c`
- The top of the .c file must include the student numbers and full names of all group members in comment lines.
- **Submission deadline:** *14 December 2025 – 23:59*
  - *Late submissions will not be accepted or graded.*
- **Submission email:** `ikchusnaibis@gmail.com`
  - *Submissions to any email address other than the above provided will not be accepted or graded.*

Demo Presentation:

- Groups will present their project on 23 December 2025, for a brief 5 to 10 minutes demo. (Further details regarding the schedule and location will be announced later.)
- All members of the group are responsible for the entire project.
- The demo presentation will be conducted using the code directly. No additional presentation slides are required. Demo will be run on the code.
- The demo is mandatory, and all group members must attend. Projects without a demo will not be evaluated, and members who do not attend will be considered as not having submitted the assignment.
- Group members who cannot explain the project or provide satisfactory answers during the demo will receive significantly lower grades.

NOTE: Projects found to contain plagiarism or copied content will not be evaluated under any circumstances. Responsibility for such issues lies entirely with the students.

Grading Scores (Total 100)

- (10 pts) The program repeatedly prompts the user to enter an ID number until a valid one is entered, and the user is allowed to exit the program at any time at this stage.
- (30 pts) The program correctly verifies the entered ID number.
- (4 pts) Once a valid ID number is entered, the program immediately proceeds to the special ID number check and prints the result.
- (10 pts) The `specialityChecker` function receives required info from `main` and returns its result to `main`. The printing of this result is correctly handled within `main`.
- (10 pts) The special ID number check is performed correctly.
- (4 pts) After a valid ID number is entered, either `seasonFinder` or `vowelCounter` is invoked correctly based on the special ID evaluation.
- (20 pts) `seasonFinder` correctly verifies the month inputs. It allows up to three entries for month, displaying remaining attempts after each invalid input. On a valid month, it prints the correct season; after three failures, it informs the user and exits.
- (8 pts) The `vowelCounter` function correctly counts the vowels and prints the results.
- (4 pts) The program terminates properly after the execution of `seasonFinder` or `vowelCounter`.

Penalty Scores (Total -75)

- (-15 pts) Use of programming constructs or features not covered between Lec01 and Lec09.
- (-10 pts) Use of the switch statement.
- (-10 pts) Use of the if-else statement.
- (-10 pts) Use of the while or do...while statements.
- (-10 pts) Use of the for loop.
- (-12 pts) Failure to define all four required functions or failure of any of them to operate as specified.
- (-8 pts) Use of meaningless function or variable names.

And additional penalty scores in demo presentation…

Example runs of program: (Text shown in **bold** represents user input; all other text is output generated by the program.)

```
Enter a Turkish ID Number (-1 to exit!): 34726519388
The entered Turkish ID number is special!
Enter the month you were born (1-12):
10
You were born in Autumn.
```

```
Enter a Turkish ID Number (-1 to exit!): 63712063514
The entered Turkish ID number is special!
Enter the month you were born (1-12):
15
Try again. You have 2 attempt(s) remaining. Enter the month you were born (1-12):
12
You were born in Winter.
```

```
Enter a Turkish ID Number (-1 to exit!): 34726519388
The entered Turkish ID number is special!
Enter the month you were born (1-12):
0
Try again. You have 2 attempt(s) remaining. Enter the month you were born (1-12):
19
Try again. You have 1 attempt(s) remaining. Enter the month you were born (1-12):
5
You were born in Spring.
```

```
Enter a Turkish ID Number (-1 to exit!): 63712063514
The entered Turkish ID number is special!
Enter the month you were born (1-12):
20
Try again. You have 2 attempt(s) remaining. Enter the month you were born (1-12):
-6
Try again. You have 1 attempt(s) remaining. Enter the month you were born (1-12):
0
No attempts left. Program terminated.
```

```
Enter a Turkish ID Number (-1 to exit!): 34726519388
The entered Turkish ID number is special!
Enter the month you were born (1-12):
20
Try again. You have 2 attempt(s) remaining. Enter the month you were born (1-12):
-6
Try again. You have 1 attempt(s) remaining. Enter the month you were born (1-12):
0
No attempts left. Program terminated.
```

```
Enter a Turkish ID Number (-1 to exit!): 57193418266
The entered Turkish ID number is not special!
Enter Full Name (with English letters only):
Ali Riza Binboga
Number of a: 3
Number of e: 0
Number of i: 3
Number of u: 0
Number of o: 1
```

```
Enter a Turkish ID Number (-1 to exit!): 24685071430
The entered Turkish ID number is not special!
Enter Full Name (with English letters only):
Sude Ozge Azimli
Number of a: 1
Number of e: 2
Number of i: 2
Number of u: 1
Number of o: 1
```

```
Enter a Turkish ID Number (-1 to exit!): 57193418266
The entered Turkish ID number is not special!
Enter Full Name (with English letters only):

Number of a: 0
Number of e: 0
Number of i: 0
Number of u: 0
Number of o: 0
```

```
Enter a Turkish ID Number (-1 to exit!): 24685071430
The entered Turkish ID number is not special!
Enter Full Name (with English letters only):
Ibrahim Deniz Olgunca
Number of a: 2
Number of e: 1
Number of i: 3
Number of u: 1
Number of o: 1
```

```
Enter a Turkish ID Number (-1 to exit!): -1
```

```
Enter a Turkish ID Number (-1 to exit!): 34726519
Invalid Turkish ID Number! Enter a Turkish ID Number (-1 to exit!): -1
```

```
Enter a Turkish ID Number (-1 to exit!): 03712663514
Invalid Turkish ID Number! Enter a Turkish ID Number (-1 to exit!): 63712053513
Invalid Turkish ID Number! Enter a Turkish ID Number (-1 to exit!): -1
```

```
Enter a Turkish ID Number (-1 to exit!): 34726519386
Invalid Turkish ID Number! Enter a Turkish ID Number (-1 to exit!): 03712663514
Invalid Turkish ID Number! Enter a Turkish ID Number (-1 to exit!): 4891245654684654
Invalid Turkish ID Number! Enter a Turkish ID Number (-1 to exit!): 63712053513
Invalid Turkish ID Number! Enter a Turkish ID Number (-1 to exit!): 34726519388
The entered Turkish ID number is special!
Enter the month you were born (1-12):
7
You were born in Summer.
```

```
Enter a Turkish ID Number (-1 to exit!): 4891245654684654
Invalid Turkish ID Number! Enter a Turkish ID Number (-1 to exit!): 0
Invalid Turkish ID Number! Enter a Turkish ID Number (-1 to exit!): -2
Invalid Turkish ID Number! Enter a Turkish ID Number (-1 to exit!): -63712063514
Invalid Turkish ID Number! Enter a Turkish ID Number (-1 to exit!): 24685071430
The entered Turkish ID number is not special!
Enter Full Name (with English letters only):
Ömer Güral Yanık
Number of a: 2
Number of e: 1
Number of i: 0
Number of u: 0
Number of o: 0
```

Student No	Name	Surname	Group
240401051	Esma	Canbaşı	Group 1
240401036	Muhammet Emin	Aras	
240401082	Esma Nur	Noyan	Group 2
250401016	Rozerin	Sun	
250401085	Eylül Tuana	Sulukaya	Group 3
250401099	Ceyda	Yıldız	
250401039	Nihal	Çelik	
250401009	Beren	Avcı	Group 4
240401072	Abdullah	Karaman	
250401015	Ege	Sonbahar	Group 5
240401037	Efe	Arslan	
250401012	Görkem	Sezen	
250401052	Abidin Aras	Güler	Group 6
250401053	Kamil Kaan	Günay	
250401079	Berkay Eren	Onaç	
250401014	Sena	Boncuk	Group 7
240401112	Yaren	Yıldırım	
240401059	Emre	Demir	Group 8
240401100	Uğur	Temel	
240401075	Aybüke	Kılavuz	Group 9
240401102	Halime Cemre	Tunçay	
240401043	Ceren	Bakışkan	
220401070	Burak	Sarı	Group 10
250401017	Eren	Çakır	
240401070	Ali	Güneyli	Group 11
240401076	Ali	Kılınç	
240401089	Sinan	Peker	Group 12
240401061	Harun	Demirdağ	
240401050	Hasan Onur	Can	Group 13
240401087	Burak	Pala	
240401049	Serdar	Büküm	
240401062	Halil İbrahim	Doymuş	Group 14
240401108	Görkem	Vuruşkan	
240401110	Bahadır	Yeniçay	
220401047	Hikmet Berke	Dindar	Group 15
250401019	Mehmet Boran	Yıldırım	
240401086	Elanur	Özcan	Group 16
240401046	Can	Beran	
250401104	Bilge	Zeybek	Group 17
240401078	Talha	Korkmaz	
240401079	Musa	Korkut	Group 18
240401114	Enis Mert	Ziya	
240401040	Yiğit	Ayakdaş	
230401059	Yusuf Kayra	Karakurt	Group 19
240401035	Arda	Aktürk	
250401037	Ezgi Dilan	Çankaya	Group 20
250401073	Ahsen Sıla	Kuru	
250401030	Sıla Yüksel	Apaydın	
230401034	Fatmanur	Bulut	Group 21
240401101	Erkin	Tilkioğlu	
240401064	Sude Nur	Eger	Group 22
240401091	Nisa Nur	Sarı	
240401052	Bertuğ Mehmet	Ceylan	Group 23
240401008	Muhammet Hüseyin	Coşkun	
240401053	Beyza	Çavaş	
240401080	Mustafa Emir	Küçük	Group 24
250401063	Hüseyin Efe	Kaya	
250401062	Kağan Can	Kaya	Group 25
250401083	Alperen	Sarı	
230401095	Mehmet Meriç	Yüksel	Group 26
230401063	Abdullah Samet	Kaşka	
230401094	Halil Kerim	Yücel	
250401106	Hamza	Aşıkoğlu	Group 27
250401107	İsmail	Kalay	
240401074	Ali	Kaymak	Group 28
250401089	Berke	Şenol	
250401095	Deniz Kağan	Uysal	
240401081	Orkun	Mutaf	Group 29
240401077	Yakup	Kıymaz	
240401047	Ayberk	Bulut	
250401092	İbrahim Alp	Turan	Group 30
250401006	Mahammad	Nazarli	
240401129	Fatımanur	Güler	Group 31
240401068	Ege	Erpul	
240401109	Zehra Erdem	Yalçınkaya	
240401083	Enes	Oral	Group 32
240401084	Emirhan	Orhan	
240401106	Mert	Ünal	Group 33
240401092	Arda Ahmet	Sevin	
240401042	Hayriye Dilvin	Aydoğan	Group 34
250401069	Murat	Koç	
240401105	Yunus Emre	Uygun	Group 35
240401045	Feyzanur	Bektaş	
230401079	Halil	Taşyürek	
230401022	Olcan Osman	Akyüz	*Group 36
230401070	Tufan	Mudarasız	
220401090	Mehmet Buğra	Yurttaş	
240401127	Berkant	Ünlü	*Group 37
210401061	Samet Kağan	Demir	
220401046	Barış	Derelioğlu	
240401057	Sultan	Demir	*Group 38
240401039	Şeyma	Asarlık	
230401089	Behzat	Yıldırım	*Group 39
230401090	Mehmetcan	Yıldırım	
230401060	Miraç	Karasu	
230401085	Ahmet Buğra	Yanga	*Group 40
230401039	Hasan	Demirci	
220401088	Mahmut	Yiğit	

Students who were not part of any group were randomly assigned to new groups, which are marked with an asterisk (*).